

Genetic report for longevity

Anxiety and Stress Resilience

Polygenic risk of anxiety and concern, stress axis (FKBP5, NR3C1, CRHR1), GABA, serotonin and NPY, trauma and sensitivity to stress

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Anxiety and how to recover from stress have a moderate genetic component (about 30%), very shared with depression and neuroticism. This report reads the polygenic scores of concern, nervousness and sensitivity, the GWAS loci of anxiety, posttraumatic stress disorder and stress sensitivity, and the genes of the hypothalamus-pituitary-suprarenal axis (FKBP5, NR3C1, CRHR1), GABA, serotonin and NPY. Neurotransmitter genes are left unbiased, as in the profile (§14, §15) and in the Dopamine and Noradrenaline reports. The scores of anxiety, panic and post-traumatic stress already calculated in the Psychodelical report are reused.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

22

variants

12

in the chip

9

imputed
($DR2 \geq 0.8$)

1

not found

10

Scores with
percentile

Overall reading. Anxiety is typical to favourable and the axis of stress is favourable; the point against is in the loci of trauma. In the UK Biobank scores: suffering from "nerves" in the 21st percentile and sensitivity to criticism in the 24 (in favor); feeling worried or anxious in the 67, worrying after an embarrassment in the 48 and depression linked to stressful events in the 33 (typical). From the Psychodelic report: generalized anxiety in 26.5, panic in 55, posttraumatic stress in 59 (low coverage). The scores of anxiety loci (60), concern (68) and stress sensitivity (54) are typical; posttraumatic stress (94 loci) is in the 84th percentile, with ESR1 T/T against and NCAM1 C/C in favor. In the axis of stress: FKBP5 rs1360780 C/C, without the T allele that makes cortisol recovery slower after trauma; genetic cortisol in the 40th percentile. In CRHR1, rs7207400 C/C (2 copies of the most concerned allele) probably marks the H2/H2 haplotype of the 17q21 inversion (see Personality report). Without judgment, as in the profile: COMT Val/Val (the "guerreiro" profile, plus stress tolerance in average), GABRA2 T/T, GAD1 G/G, NPY T/T, RGS2 C/G, NR3C1 Bcll C/G and 9 β T/T. Caffeine-related anxiety is "less likely" (ADORA2A C/C, Health Report).

Favourable

- **Suffering from "nerves" (3348 variants):** the 21st percentile
- **Sensitivity / feeling hurt (7922 variants):** the 24th percentile
- **Caffeine-linked anxiety ("less likely"):** — (Health Report)
- **FKBP5 rs1360780 CC:** FKBP5 rs1360780: the minor allele (T) increases the induction of FKBP5 by cortisol and the recovery of stress is slower; the effect appears mainly with early trauma.
- **DENND1B rs2488401 TC:** DENND1B: locus of anxiety and concern.
- **NCAM1 rs7106434 CC:** NCAM1: neuronal adhesion; posttraumatic stress locus.

Unfavourable

- **Posttraumatic stress (GWAS loci):** the 84th percentile
- **CRHR1 rs7207400 CC:** CRHR1 (17q21): locus of concern and neuroticism.
- **ESR1 rs9479138 TT:** ESR1: estrogen receptor alpha.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 12 chip genotypes, 0 needed complement, 2 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand — such as COL1A1 or VDR).

Anxiety, worry and nervousness

Scores of UK Biobank (feel worried or anxious, worry after an embarrassment, "nerves", sensitivity to criticism) and loci of the GWAS of anxiety and concern. Generalized anxiety and panic come from the Psychedelic report.

Ageing mechanisms: Intercellular communication alteration; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Feeling worried or anxious (9747 variants)	Typical / intermediate	the 67th percentile · z=0.41	Higher score : more concern. PGS001021 (PGS Catalog; snpnet; Feelings of worry or anxiety; PGP000244, PMID 35324888). Variants with genotype: 8626 of 9747 (covering of 89.9% coefficients).
Concern long after an embarrassment (7714 variants)	Typical / intermediate	the 48th percentile · z=-0.09	Higher score = more rumination. PGS001022 (PGS Catalog; snpnet; Worry too long after an entanglement experience; PGP000244, PMID 35324888). Variants with genotype: 6790 of 7714 (covering of 88.7% coefficients).
Suffering from "nerves" (3348 variants)	Favourable	the 21st percentile · z=-0.86	Higher score : more nervousness. PGS001017 (PGS Catalog; snpnet; Suffer from 'nerves'; PGP000244, PMID 35324888). Variants with genotype: 2925 of 3347 (covering of 88.5% coefficients).
Sensitivity / feeling hurt (7922 variants)	Favourable	the 24th percentile · z=-0.72	Higher score = more emotional sensitivity. PGS001016 (PGS Catalog; snpnet; Sensitivity/hurt feelings; PGP000244, PMID 35324888). Variants with genotype: 6977 of 7921 (covering of 89.0% coefficients).
Anxiety (loci of GWAS)	Typical / intermediate	the 60th percentile · z=0.25	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 89 of 92 loci with genotype. Studies: Frilighou E (Anxiety disorder); Li W (Anxiety disorder); Otowa T (Anxiety disorder); Thorp JG (Anxiety); Wendt FR (Generalized anxiety disorder) Wendt FR (Generalized anxiety disorder)). Variants with genotype: 89 of 92 (covering of 96.7% coefficients).
Concern (loci of GWAS)	Typical / intermediate	the 68th percentile · z=0.52	Higher score : more concern. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 91 of 91 loci with genotype. Studies: Nagel M (Feeling worry); Nagel M (Worry). Variants with genotype: 91 of 91 (covering of 100.0% coefficients).
Generalized anxiety (PGS018941)	Typical / intermediate	the 26th percentile	Typical likelihood label; score developed in another population, uncertain transferability. Source: Psychedelic report.
Panic attacks (PGS004451)	Typical / intermediate	the 55th percentile	It labels "typical likelihood". Source: Psychedelic report.

Favourable

- Suffering from "nerves" (3348 variants) (the 21st percentile)
- Sensitivity / feeling hurt (7922 variants) (the 24th percentile)
- DENND1B rs2488401 TC

Unfavourable

- CRHR1 rs7207400 CC

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DENND1B	rs2488401 1:197733271 · C/A/T	TC	imputation DR2=1.00	C (1; 82%)	moderate	DENND1B: locus of anxiety and concern. Meaning: GWAS Catalog: Anxiety (Thorp JG 2021; p=8e-19); C ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CRHR1	rs7207400 17:45746994 · T/C	CC	imputation DR2=1.00	C (2; 29%)	moderate	CRHR1 (17q21): locus of concern and neuroticism. Meaning: GWAS Catalog: Worry (Nagel M 2018; p=2e-15); allele C ↑

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
L3MBTL2	rs2273085 22:41219372 · C/T	CT	chip	T (1; 31%)	moderate	L3MBTL2: locus of anxiety in the 22 chromosome. Meaning: GWAS Catalog: Anxiety (Thorp JG 2021; p=1e-37); C ↓ allele
PTCH1	rs28427480 9:95502660 · A/C	AA	imputation DR2=1.00	C (0; 10%)	moderate	PTCH1: via Hedgehog; locus of anxiety and stress sensitivity. Meaning: GWAS Catalog: Anxiety (Thorp JG 2021; p=1e-25); Allele A ↓
CADM2	rs9854869 3:85371574 · C/A	CC	imputation DR2=0.99	C (2; 77%)	moderate	CADM2: neuronal adhesion; locus of concern and risk. Meaning: GWAS Catalog: Worry (Nagel M 2018; p=1e-16); Allele A ↓

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TMEM132D	rs7309727 12:129470814 · T/A/C	CT	chip	—	weak	TMEM132D: locus of panic disorder. No sense usable in GWAS Catalog; informative.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GRM8	rs13226841 — · —	—	SNP not found in the data	—	moderate	GRM8: glutamate metabotropic receptor. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Stress axis and cortisol

FKBP5 stops the glucocorticoid receptor (NR3C1); with early stress, the risk allele loses methylation and the axis becomes more reactive. CRHR1 is the corticotropin-releasing hormone receptor; SERPINA6 carries cortisol in the blood.

Ageing mechanisms: Endocrine dysregulation; Epigenetic Changes; Circadian disorders

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Cortisol in blood	No judgement (informative)	the 40th percentile · $z=-0.35$	Higher score = higher cortisol. Information; few loci. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; $\ln(OR)$ or beta coefficients): 6 of 6 loci with genotype. Studies: Chan II (Plasma cortisol levels (morning)); Neumann A (Cortisol levels) Pott J (Cortisol levels (plasma)). Variants with genotype: 6 of 6 (covering of 100.0% coefficients).

Favourable

- FKBP5 rs1360780 CC

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FKBP5	rs1360780 6:35639794 · T/A/C	CC	chip	T (0; 31%)	moderate	FKBP5 rs1360780: the minor allele (T) increases the induction of FKBP5 by cortisol and the recovery of stress is slower; the effect appears mainly with early trauma. C/C: without the T-risk allele.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NR3C1 (BclI)	rs41423247 5:143399010 · G/C	CG	chip	—	weak	NR3C1 BclI: G allele is associated with more sensitivity to glucocorticoids.
NR3C1 (9β)	rs6198 5:143278056 · T/C	TT	imputation DR2=1.00	—	weak	NR3C1 9β A3669G: the G allele stabilizes the β isoform and gives some resistance to glucocorticoids. T/ T = without the G allele (9β).
CRHR1	rs110402 17:45802681 · G/A/C	GG	chip	—	weak	CRHR1: allele A was associated with protection against depression after childhood maltreatment (gene x environment interaction).
ADCYAP1R1	rs2267735 7:31095890 · C/G	CG	chip	—	weak	ADCYAP1R1 (PACAP receiver): C allele was associated with posttraumatic stress, especially in women.

GABA, serotonin, NPY and catecholamines

GABRA2 and GAD1 (GABA), NPY (Y peptide, linked to resilience), RGS2, COMT. No good/bad judgment, as in the profile (§14, §15, §21).

Ageing mechanisms: Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Caffeine-linked anxiety ("less likely")	Favourable	—	It labels "less likely" (ADORA2A C/C). Source: Health Report.

Favourable

Unfavourable

• Caffeine-linked anxiety ("less probable")

None.

Non-judgmental variants (informative) (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GABRA2	rs279858 4:46312576 · T/A/C	TT	chip	—	weak	GABRA2 (α2 subunit of the receptor GABA-A): anxiety and response to benzodiazepines; without judgment, as in §14.
GAD1	rs3749034 2:170816965 · G/A/C/T	GG	chip	—	weak	GAD1: synthesis of GABA; without judgment, as in §14.
NPY	rs16147 7:24283791 · T/A/C	TT	chip	—	weak	NPY -399T>C: expression of Y peptide (NPY), linked to stress resilience.
RGS2	rs4606 1:192812042 · C/G/T	CG	chip	—	weak	RGS2: regulates G protein-coupled receptors; associated with anxiety in small studies.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Val/Val is the "warrior" profile (more tolerance to stress and pain, less anxiety on average); without judgment, as in §21. G/G = Val/Val.

Trauma and stress sensitivity

Loci of posttraumatic stress (FOXP2, NCAM1, GABBR1, ESR1) and sensitivity to stress and adversity; score of depression linked to stressful events.

Ageing mechanisms: Neuroinflammation; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Stress-related depression (545 334 variants)	Typical / intermediate	the 33rd percentile · z=-0.44	Higher score : higher risk. PGS002113 (PGS Catalog; LDpred2 (bigsnp); Depression was possibly related to stressful or traumatic event; PGP000263, PMID 34995502). Variants with genotype: 533880 of 545334 (covering of 97.0% coefficients).
Posttraumatic stress (GWAS loci)	Unfavourable	the 84th percentile · z=1.04	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 94 of 97 loci with genotype. Studies: Maihofer AX (Posttraumatic stress disorder); Nievergelt CM (Posttraumatic stress disorder); Stein MB (Posttraumatic stress disorder). Variants with genotype: 94 of 97 (covering of 96.9% coefficients).
Stress and adversity sensitivity	Typical / intermediate	the 54th percentile · z=0.07	Higher score = more sensitivity. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 46 of 46 loci with genotype. Studies: Nagel M (Sensitivity to environmental stress and adversity). Variants with genotype: 46 of 46 (covering of 100.0% coefficients).
Posttraumatic stress (PGS005393; 34% coverage)	Typical / intermediate	the 59th percentile	Typical likelihood label; low coverage. Source: Psychedelic report.

Favourable

- NCAM1 rs7106434 CC

Unfavourable

- Posttraumatic stress (GWAS loci) (the 84th percentile)
- ESR1 rs9479138 TT

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NCAM1	rs7106434 11:112989857 · T/C	CC	imputation DR2=0.99	T (0; 45%)	moderate	NCAM1: neuronal adhesion; posttraumatic stress locus. Meaning: GWAS Catalog: Posttraumatic stress disorder (Nievergelt CM 2024; p=4e-16); T ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ESR1	rs9479138 6:151894064 · T/A/G	TT	imputation DR2=1.00	T (2; 32%)	moderate	ESR1: estrogen receptor alpha. Meaning: GWAS Catalog: Posttraumatic stress disorder (Nievergelt CM 2024; p=3e-13); T ↑ allele

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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FOXP2	rs1476535 7:114430980 · C/A/G/T	CT	imputation DR2=0.98	T (1; 50%)	moderate	FOXP2: transcription factor; locus of posttraumatic stress and ADHD. Meaning: GWAS Catalog: Posttraumatic stress disorder (Nievergelt CM 2024; p=4e-20); T ↑ allele
GABBR1	rs28986300 6:29583119 · A/G	AA	imputation DR2=1.00	G (0; 6%)	moderate	GABBR1: GABA-B receiver. Meaning: GWAS Catalog: Posttraumatic stress disorder (Nievergelt CM 2024; p=8e-14); allele A ↓
CELF4	rs11665070 18:37572600 · G/A	AG	chip	G (1; 34%)	moderate	CELF4: regulation of neuronal RNA; sensitivity to stress and well-being. Meaning: GWAS Catalog: Sensitivity to environmental stress and adversity (Nagel M 2020; p=5e-18); G ↑ allele

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Neuroinflammation	CRHR1 CC; Posttraumatic stress (GWAS loci) (the 84th percentile); ESR1 TT	To suffer from "nerves" (3348 variants) (the 21st percentile); Sensitivity / feeling hurt (7922 variants) (the 24th percentile); DENND1B TC; NCAM1 CC	To feel worried or anxious (9747 variants) (the 67th percentile); Concern long after an embarrassment (7714 variants) (the 48th percentile); Anxiety (loci of GWAS) (percentile 60); Concern (loci of GWAS) (percentile 68); Generalized anxiety (PGS018941) (the 26th percentile); Panic attacks (PGS004451) (the 55th percentile); L3MBTL2 CT; PTCH1 AA
Chronic Inflammation	Posttraumatic stress (GWAS loci) (the 84th percentile); ESR1 TT	NCAM1 CC	Stress-related depression (545 334 variants) (the 33rd percentile); Stress and adversity sensitivity (the 54th percentile); Posttraumatic stress (PGS005393; 34% coverage) (the 59th percentile); FOXP2 CT; GABBR1 AA; CELF4 AG
Intercellular Communication	CRHR1 CC	To suffer from "nerves" (3348 variants) (the 21st percentile); Sensitivity / feeling hurt (7922 variants) (the 24th percentile); DENND1B TC; Caffeine-linked anxiety ("less probable")	To feel worried or anxious (9747 variants) (the 67th percentile); Concern long after an embarrassment (7714 variants) (the 48th percentile); Anxiety (loci of GWAS) (percentile 60); Concern (loci of GWAS) (percentile 68); Generalized anxiety (PGS018941) (the 26th percentile); Panic attacks (PGS004451) (the 55th percentile); L3MBTL2 CT; PTCH1 AA
Epigenetic Changes	—	FKBP5 CC	—
Endocrine dysregulation	—	FKBP5 CC	—
Circadian disorders	—	FKBP5 CC	—

Practical part

Ideas for anxiety and stress, crossed with the genetic plan v2 and current medication (isotretinoin, lithium 5 mg, low dose naltrexone 4.5 mg, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin). The current stack remains as it is; everything is suggestion (not applied). The genetics of the stress axis is favourable; the objective is to keep the recovery fast and not let the concern grow.

1. Slow breathing 5-10 minutes per day

Genetics: Concern p67-p68; FKBP5 C/C

Breathing at ~6 cycles per minute (inspire 4–5 s, exhale 5–6 s) activates the vagus and reduces the activation of stress in a few minutes. Useful before sleep and in moments of tension.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

2. Regular exercise as anxiolytic

Genetics: Typical anxiety; posttraumatic stress p84 (loci)

Aerobic and strength exercise reduces anxiety in trials; maintaining the plan routine and not cutting it in periods of more stress.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Tools for rumination (suggestion)

Genetics: Concern after an embarrassment p48; concern p67

If concern or stress rises: cognitive-behavioral therapy techniques or mindfulness (8 weeks) have the best evidence; preferable to any supplement.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

4. Caffeine within the limit and L-theanine

Genetics: ADORA2A C/C; caffeine-related anxiety "less likely"

Caffeine is well tolerated, but keep up to 200 mg only in the morning. L-theanine 100–200 mg in stress days is suggested (small tests).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

5. Magnesium and sleep

Genetics: Insomnia p89 (Circadian Rhythm and Sleep Report)

Magnesium glycinate already taken and a fixed sleep time help to reduce nocturnal activation; melatonin 0.3–1 mg (the dose of the plan).

V2 genetic plan and medication: Already in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Benzodiazepines of regular use (tolerance and dependence; GABRA2 only informative). *suggestion (not applied)*
- Ashwagandha and kava: in conflict with the plan (liver). *suggestion (not applied)*
- Rhodiola without the doctor (CYP3A4 and CYP2C9; see report Dopamine and Noradrenaline). *suggestion (not applied)*
- Stimulants or caffeine in the afternoon on stress days. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found.
- The UK Biobank scores come from simple questions (yes/no); they measure trends, not diagnosed disturbances.
- The effects of FKBP5, CRHR1 and NR3C1 depend on the environment (early trauma) and do not read alone.
- Neurotransmitter genes are left unbiased, as in the profile (§14, §15, §21).
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001021 (PGS Catalog; snpnet; Feelings of worry or anxiety; PGP000244, PMID 35324888 · PGS001022 (PGS Catalog; snpnet; Worry too long after an entanglement experience; PGP000244, PMID 35324888 · PGS001017 (PGS Catalog; snpnet; Suffer from 'nerves'; PGP000244, PMID 35324888 · PGS001016 (PGS Catalog; snpnet; Sensitivity/hurt feelings; PGP000244, PMID 35324888 · PGS002113 (PGS Catalog; LDpred2 (bigsnpr); Depression was possibly related to stressful or traumatic event; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentis: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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